

SPARTA, IL, USA

WMO: 744653

Lat: 38.149N

Lon: 89.699W

Elev: 538

StdP: 14.41

Time Zone: -6.00 (NAC)

Period: 02-19

WBAN: 63814

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	9.4	13.9	-2.1	5.0	13.9	2.4	6.4	16.8	24.0	45.1	21.0	41.2	6.1	310	0.440

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	17.9	93.7	76.3	91.3	76.3	90.0	75.8	81.2	88.2	79.5	86.9	78.0	86.0	7.2	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
79.2	154.9	85.2	77.3	144.8	84.0	75.3	135.5	82.5	45.2	88.1	43.2	86.3	41.9	86.3	87.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
19.9	17.6	15.8	3.3	98.6	4.0	3.4	0.4	101.0	-2.0	103.0	-4.2	104.9	-7.1	107.4		
			1.7	82.8	4.4	2.5	-1.5	84.6	-4.1	86.0	-6.6	87.4	-9.8	89.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	57.1	32.9	36.7	47.7	57.4	66.8	75.9	78.4	76.8	70.0	58.4	46.3	37.3
	DBStd	18.23	11.53	11.25	10.84	9.06	8.09	5.65	5.45	5.41	6.74	9.26	9.75	9.93
	HDD50	1760	543	393	177	34	2	0	0	0	0	24	180	407
	HDD65	4372	996	793	544	255	81	4	0	1	29	249	562	858
	CDD50	4369	12	20	105	256	521	776	879	829	601	285	70	15
	CDD65	1506	0	0	6	27	136	329	414	366	180	45	3	0
	CDH74	14292	0	0	32	209	1066	3257	4337	3437	1578	363	13	0
	CDH80	5560	0	0	1	26	280	1305	1902	1402	551	92	1	0
Wind	WSAvg	6.4	7.9	7.6	8.1	8.0	6.3	5.4	4.7	4.2	4.4	5.9	7.1	7.4
Precipitation	PrecAvg	42.4	2.3	2.4	4.1	4.2	4.6	3.8	4.0	3.3	3.0	3.0	4.0	3.1
	PrecMax	54.4	7.6	4.4	9.6	11.3	12.1	8.9	12.9	9.1	8.2	7.5	11.5	10.6
	PrecMin	27.3	0.2	0.3	1.1	1.2	0.9	0.3	0.4	0.3	0.2	0.0	0.6	0.7
	PrecStd	6.9	1.7	1.2	2.0	2.4	2.6	2.2	2.5	1.9	1.8	1.7	2.4	2.1
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	65.9	70.0	78.6	82.3	90.1	95.4	99.4	98.9	92.8	87.8	75.0	65.6
		MCWB	61.7	61.5	64.7	66.4	73.1	75.6	75.8	74.8	72.2	72.7	62.2	61.7
	2%	DB	62.5	64.4	73.0	80.3	86.0	91.8	94.7	93.2	89.8	82.1	70.0	61.4
		MCWB	59.1	57.3	62.3	65.5	72.7	76.1	77.5	75.5	73.4	68.4	59.8	56.9
	5%	DB	55.1	60.1	69.7	75.5	82.4	90.2	91.4	90.4	86.2	78.5	65.6	56.8
		MCWB	50.6	53.9	59.8	63.4	70.5	75.4	77.7	76.0	72.8	65.4	58.6	52.1
	10%	DB	49.5	54.6	64.4	72.7	80.8	87.9	89.8	87.9	82.4	73.3	62.6	52.2
		MCWB	45.3	48.5	57.2	62.1	69.6	74.1	77.2	75.7	69.8	63.6	55.5	48.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	63.5	63.9	67.4	72.0	77.5	81.3	84.2	83.2	78.6	74.3	66.8	62.9
		MCDB	64.4	67.4	74.3	78.2	85.0	88.4	91.6	89.6	84.8	83.4	72.6	64.7
	2%	WB	59.0	59.6	64.3	68.4	75.3	79.2	81.6	80.7	76.9	71.7	63.2	58.0
		MCDB	62.2	63.6	71.1	76.3	82.1	88.1	89.0	87.4	85.1	78.9	66.4	61.3
	5%	WB	51.8	54.8	61.8	66.0	73.3	77.7	80.0	78.8	75.0	69.0	59.8	53.5
		MCDB	54.6	58.7	67.9	72.5	80.1	86.7	87.3	86.2	82.4	75.1	64.8	55.7
	10%	WB	45.0	48.8	58.8	63.7	71.3	76.3	78.3	77.2	73.1	65.9	55.8	48.2
		MCDB	48.7	53.8	63.9	69.6	78.0	84.5	86.0	85.0	79.4	71.7	61.0	51.1
Mean Daily Temperature Range	5% DB	MDBR	15.1	16.1	17.4	19.2	18.1	18.6	17.9	18.4	20.3	19.6	17.1	14.0
		MCDBR	21.6	24.0	21.9	22.7	20.2	20.6	20.0	21.1	23.5	23.3	21.3	18.9
		MCWBR	17.5	18.0	14.3	12.7	10.0	8.3	8.0	8.9	9.8	11.3	14.5	15.4
	5% WB	MCDBR	19.4	21.7	19.5	19.4	17.5	18.0	18.2	17.9	19.0	19.8	18.9	17.1
		MCWBR	18.3	18.3	14.2	13.0	9.9	8.8	9.1	9.4	9.7	11.8	15.8	16.0
Clear-Sky Solar Irradiance	taub	0.308	0.321	0.352	0.397	0.430	0.441	0.466	0.463	0.405	0.352	0.315	0.302	
	taud	2.472	2.421	2.396	2.267	2.234	2.227	2.165	2.167	2.317	2.464	2.509	2.499	
	Ebn at Noon	278	288	288	280	271	267	259	257	267	273	273	272	
	Edh at Noon	26	30	35	42	44	44	47	46	37	29	24	23	
All-Sky Solar Radiation	RadAvg	660	881	1197	1577	1771	2035	1972	1789	1503	1102	746	553	
	RadStd	70	99	114	96	155	133	130	99	95	116	81	55	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (33 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+140

Nomenclature: See separate page